

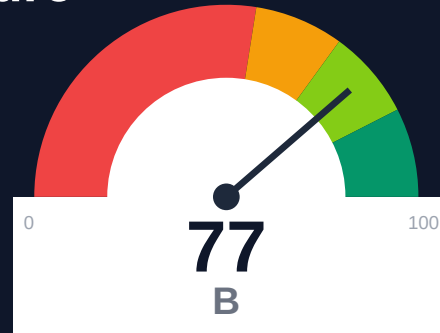
Windmill

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: Moderate

Key Metrics

Pages scanned: 973

Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 3
- SEO/GEO issue rate is high: 22.43%
- API usage-doc coverage is low: 66.67% (86 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit of windmill.dev/docs crawled 973 pages with 100% coverage against discovered scope. Site contains 3 confirmed broken internal links within user-facing documentation. API reference coverage stands at 66.67% across 86 detected API pages. SEO/geo issue rate is elevated at 22.43%, and only 53.65% of pages display last-updated metadata. Example code reliability estimate is 83.09% with overall audit confidence at 91.5%.

External posture: SEO/GEO issue rate **22.4%**, public API coverage **66.7%**, broken links **3**.

77 Audit Score	B Grade	Moderate Risk Band
973 Pages Scanned	3 Broken Links	22.4% SEO Issue Rate

Per-Site Breakdown

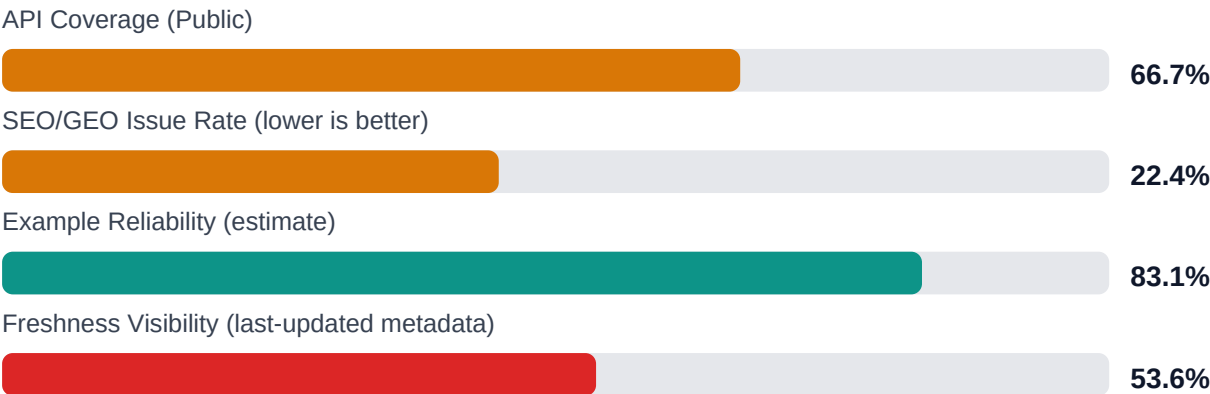
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://windmill.dev/docs	973	3	66.7	83.1	22.4

Industry Benchmark Comparison

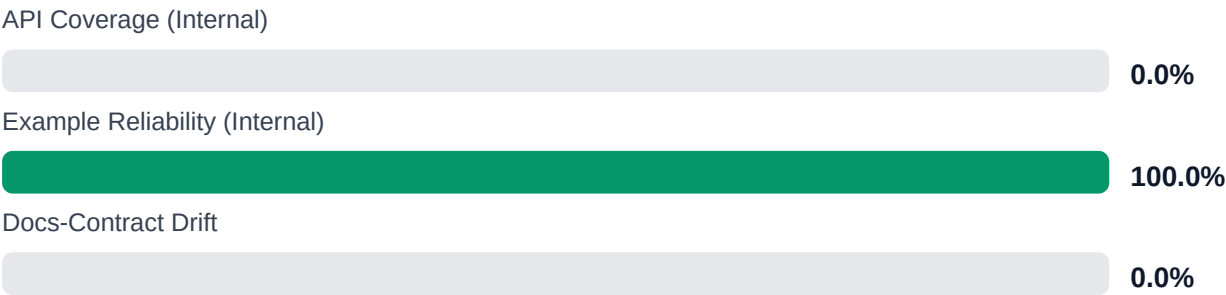
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-15
Industry average (developer docs)	85	-8
Minimum acceptable	70	+7
Your score	77	-

Gap to industry average: **8 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure	\$249,513 annual exposure	\$431,728 annual exposure
Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix 3 confirmed broken internal documentation links immediately [impact: high, effort: low]
2. Complete API reference documentation for the missing 33.33% of detected API surface (29 pages) [impact: critical, effort: high]

3. Add last-updated timestamps to 450+ pages missing freshness metadata [impact: medium, effort: medium]
4. Remediate SEO/geo issues affecting 218+ pages (22.43% of site) [impact: high, effort: medium]
5. Audit and fix example code reliability issues in the 16.91% flagged subset [impact: high, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage (100%) with 973 pages successfully examined
- + High overall audit confidence at 91.5% with strong link verification (95% confidence)
- + No repository navigation link breakage (repo_broken_links_count: 0)
- + Example code reliability estimate above 80% (83.09%)
- + Single-product topology provides clear documentation structure

Risks

- 3 confirmed broken internal links in user-facing documentation degrade user experience
- API reference coverage gap at 66.67% leaves one-third of API surface undocumented
- SEO/geo issue rate of 22.43% affects discoverability for 218+ pages
- Freshness metadata missing on 46.35% of pages (450+ pages) obscures content currency
- 16.91% of examples may contain reliability issues affecting developer onboarding
- 130 links excluded from verification may hide additional broken references
- API detection confidence at 85% suggests possible undercounted API surface

Limitations

- * External audit cannot verify accuracy or completeness of technical content, only structural and linking integrity
- * API coverage detection at 85% confidence may undercount actual API surface; manual verification recommended for complete inventory
- * Example reliability is estimated through static analysis; runtime validation and environment-specific testing not performed
- * 130 excluded links were not verified and may contain additional broken references outside audit scope
- * Freshness metadata absence does not confirm content is outdated, only that update tracking is not visible to users

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.